BG77xA-GL&BG95xA-GL
LwM2M Application Note

LPWA Module Series

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About the Document

Revision History

Version	Date	Author	Description
-	2020-11-05	Vladan RAKIC	Creation of the document
1.0	2022-09-09	Vladan RAKIC	First official release

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1 Introduction

With the rise of the Internet of Things, more and more terminal devices are being connected to the Internet, making the device management and upgrade requirements increasingly urgent. This industry challenge has been addressed with Open Mobile Alliance (OMA) Lightweight M2M (LwM2M) protocol.

LwM2M supports some features that help vendors manage their devices, update firmware, and control devices remotely, etc. Since IoT devices for embedded terminals are often very resource-constrained, without UI, and come with limited computing and network communication capabilities, the main motivation for LwM2M was to define a set of lightweight protocols for a variety of IoT devices. In addition, due to the huge number of IoT terminals, saving network resources has become increasingly important.

This document explains OMA LwM2M protocol, the architecture of LwM2M feature and how to use it on Quectel BG77xA-GL and BG95xA-GL modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Model
BG77xA-GL	BG770A-GL
	BG773A-GL
BG95xA-GL	BG950A-GL
	BG951A-GL
	BG953A-GL
	BG955A-GL

1.2. Special Mark

Table 2: Special Mark

Mark	Definition
*	Unless otherwise specified, an asterisk (*) after a function, feature, interface, pin name, AT command, or argument, indicates that the function, feature, interface, pin, AT command, or argument is under development and currently not supported; and the asterisk (*) after a model indicates that the sample of such model is currently unavailable.

2 General Overview of LwM2M

OMA LwM2M is the application layer communication protocol between an LwM2M Server and an LwM2M Client, located in an LwM2M device. The OMA LwM2M enabler includes device management and service enablement for LwM2M devices.

The modules provide the LwM2M Client on APPS. The LwM2M Client is compliant with OMA specifications and supports the following interfaces:

- Bootstrap
- Client Registration
- Device Management and Service Enablement
- Information Reporting

.

2.1. LwM2M Interfaces

2.1.1. Bootstrap Interface

Bootstrap Interface is used to provision LwM2M Client with essential information to enable it to perform the "Register" operation with one or more LwM2M Servers.

- **Bootstrap Modes**

LwM2M enabler supports four bootstrap modes: Factory Bootstrap, Bootstrap from Smartcard, Client-Initiated Bootstrap and Server-Initiated Bootstrap.

The LwM2M Client must support at least one bootstrap mode specified in the Bootstrap Interface. The modules support two bootstrap modes: Factory Bootstrap and Client-Initiated Bootstrap.

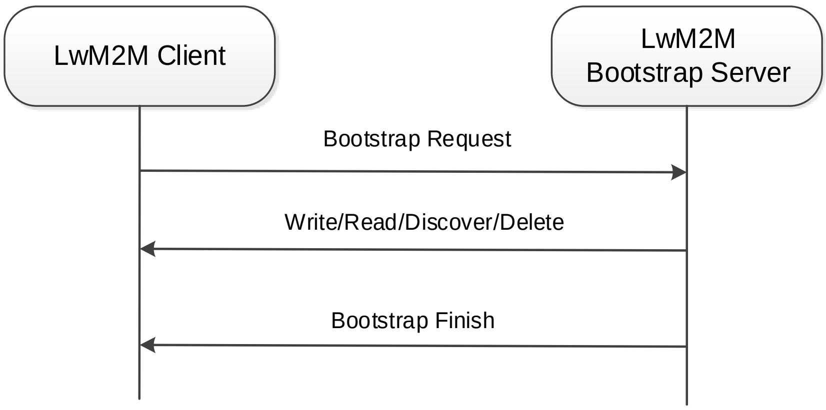

Figure 1: Operation Model for Client-Initiated Bootstrap Interface

- **Server and Access Control Configurations**

Information about LwM2M bootstrap server or other standard servers, and the access right for performing an operation for these LwM2M Servers.

2.1.2. Client Registration Interface

Client Registration Interface is used by an LwM2M Client to register on one or more LwM2M Servers, maintain each registration, and de-register from an LwM2M Server.

- **Registration**

When registering, the LwM2M Client performs the “Register” operation and provides the properties the LwM2M Server requires to contact the LwM2M Client (e.g., End Point Name); maintain the registration and session (e.g., lifetime, Queue Mode) between the LwM2M Client and LwM2M Server, as well as the knowledge of the Objects supported by LwM2M Client and existing Object Instances in the LwM2M Client. The registration is in soft state, with the lifetime indicated by the Lifetime Resource of LwM2M Server Object Instance.

- **Update**

The LwM2M Client periodically updates its registration information with the registered LwM2M Server(s) by performing the “Update” operation.

If the lifetime of a registration with an LwM2M Server expires without an update from the LwM2M Client, the LwM2M Server removes the registration of that Client.

The LwM2M Client performs the “Update” operation at an interval equal to the registration lifetime (as per Resource /1/x/1 value).

- **De-registration**

When shutting down or discontinuing the use of a LwM2M Server, the LwM2M Client performs a “De-register” operation.

The Binding Resource of the LwM2M Server Object informs the LwM2M Client of the transport protocol preferences of the LwM2M Server for the communication session between the LwM2M Client and LwM2M Server. The LwM2M Client should perform the operations in the modes indicated by the Binding Resource of the LwM2M Server Object Instance.

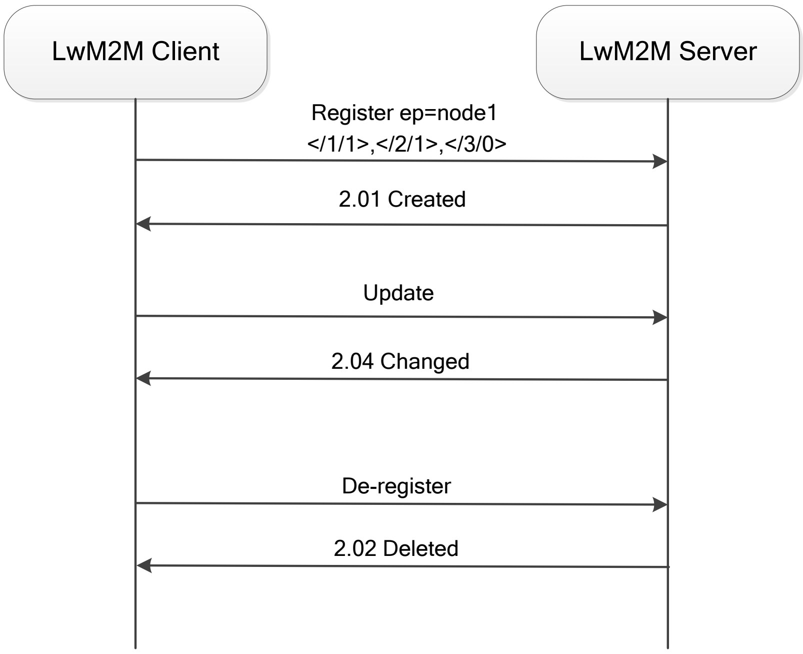

Figure 2: Example Flow for Client Registration Interface

2.1.3. Device Management and Service Enablement Interface

Device Management and Service Enablement is a very important interface in LwM2M.

- The interface is used by the LwM2M Server to access Object Instances and Resources available from a registered LwM2M Client.
- The interface provides this access through “Create”, “Read”, “Write”, “Delete”, “Execute”, “Write Attributes”, or “Discover” operations.
- The operations supported by Resource are defined in the Object definition using the Object Template.

Example flow of Device Management and Service Enablement Interface is presented below:

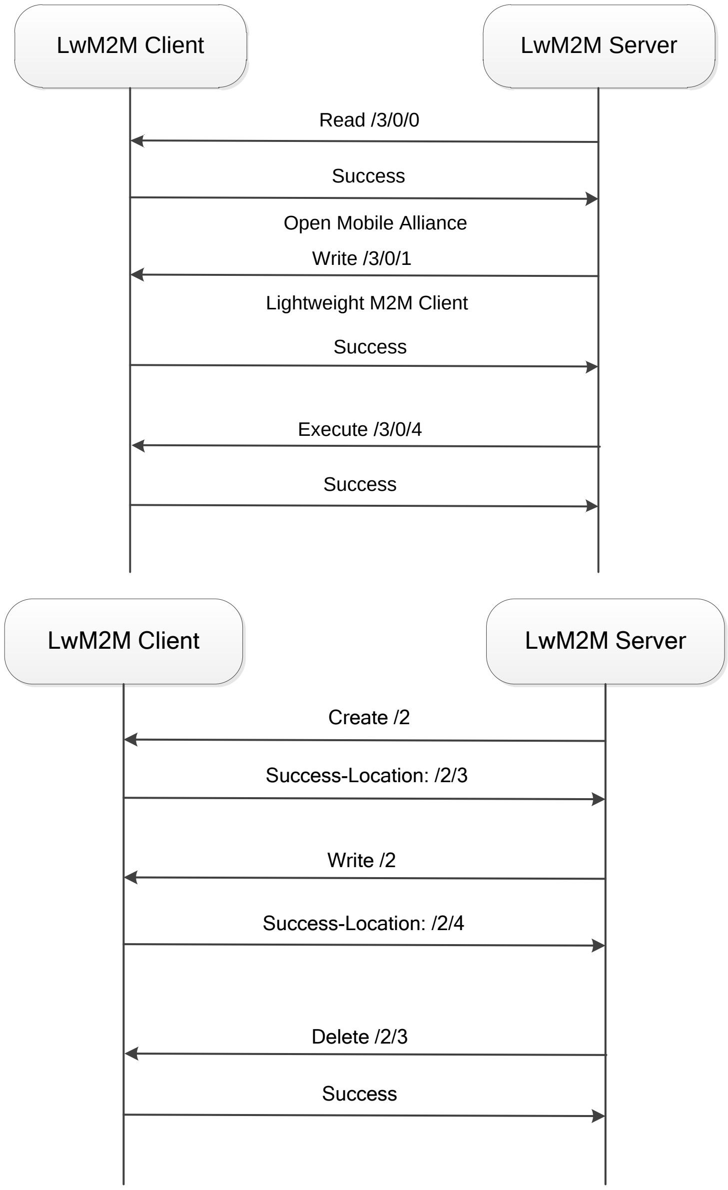

Figure 3: Example Flow of Device Management and Service Enablement Interface

2.1.4. Information Reporting Interface

The Information Reporting Interface is used by an LwM2M Server to observe any changes in a Resource on a registered LwM2M Client, and to receive notifications when new values are available.

- This observation relationship is initiated by sending an "Observe" operation to the LwM2M Client for an Object, an Object Instance, or a Resource.
- An observation ends when a "Cancel Observation" operation is performed.

LwM2M Client supports observation and notification of objects, object instances and resources.

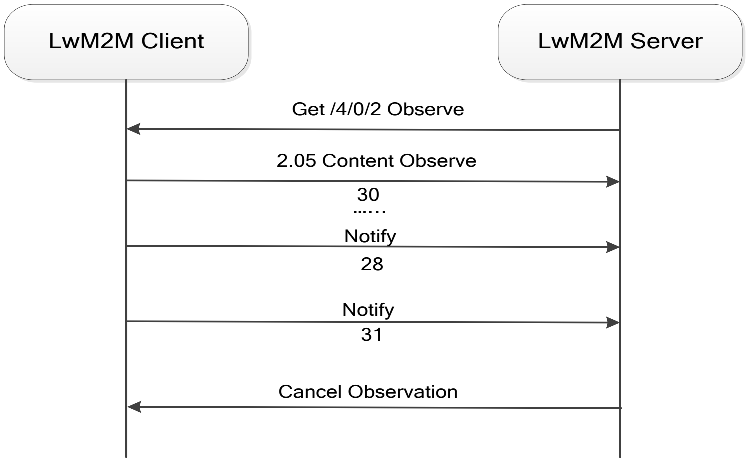

Figure 4: Example Flow of Information Reporting Interface

2.2. LwM2M Objects

The LwM2M Client of the modules implements the following objects and handles the server requests for them:

- Security object
- Server object
- Access control object
- Device object
- Connectivity monitoring object
- Location object
- Connectivity statistics object
- APN connection profile
- Cellular connectivity
- Firmware update object
- Software management object*
- Device capability management object

The information received for these objects is sent to other registered applications, such as firmware over-the-air (FOTA), which implements these objects and monitors them with LwM2M API.

For example, the LwM2M Client receives a request for an "execute update" on a firmware update object. The LwM2M Client passes the request to the FOTA application registered to receive events. When the FOTA application receives the "execute update" message, it processes and applies the image and sends the result to the LwM2M Client to forward it to the LwM2M Server.

2.3. IPv6 to IPv4 Fallback Mechanism

If IPv4v6 is selected as the IP family:

- An IPv6 call is brought up first.
- If the IPv6 call is not brought up even after all data call retries, then the LwM2M Client only attempts to bring up an IPv4 call.

If the module is given both IPv4 and IPv6 addresses:

- The module first uses the IPv6 address to connect to the LwM2M Server.
- If the LwM2M Server is unreachable with the IPv6 address, the module falls back to the IPv4 address and tries to connect to the LwM2M Server.

2.4. LwM2M and PSM

- **LwM2M Client Behavior**

The LwM2M Client must send a “registration update” to the LwM2M Server before the lifetime registration expires per the OMA specification. With the PSM enabled, the modules periodically enter deep sleep and wake up only after the PSM timer expires. During deep sleep, the LwM2M Client is shut down and cannot send the “registration update” to the server, resulting in the loss of LwM2M Client’s registration with the server.

- **How to Keep LwM2M Client Registered?***

Enable the *REG_UPDATE_ON_RECONNECT* option (see ***Chapter 0***) for the LwM2M Client to send a “registration update” to the server every time the module wakes up from PSM. Enabling this feature ensures that the LwM2M Client does not lose its registration with the server.

NOTE

Currently the modules only support enabling LwM2M PSM feature by provisioning profiles (see ***Chapter 2.5***).

2.5. File Provisioning and Configuration

2.5.1. Configuration and Script Files*

The LwM2M Client has three configuration files that can be used to control its behavior, and one script for auto starting the LwM2M Client. The four files are listed in the table below.

Table 3: Configuration and Script Files of LwM2M Client

Filename	Description
bootstrap.ini	• Used as factory bootstrap to discover the bootstrap server or other servers. • Bootstrap server details for client-initiated bootstrap. • Can be used to instantiate multiple instances of objects (Security, Server, ACL).
carrier_apn_cfg	Contains APN information for connecting to servers.
lwm2m_app_autostart	An empty file; the presence of this file indicates that the LwM2M Client can start automatically.
lwm2m_cfg	Contains LwM2M options to control features such as registration retry.

NOTE

If the above configuration files are preloaded to the */datatx* directory of the modules, the LwM2M Client starts up automatically after module reboots.

2.5.1.1. bootstrap.ini

bootstrap.ini contains the object information required for bootstrapping. The object information is in JSON format defined in the OMA specification.

Table 4: bootstrap.ini Format

Attributes	JSON Variable	Mandatory	Description
Base Name	bn	No	Base name string prepended to the Name value of the entry for forming a unique identifier for resource.
Base Time	bt	No	Base time which the Time values are relative to.

Resource Array	e Array Parameters		Yes	Resource list as JSON value array according to [SenML] with Array parameter extension (Object Link).
	Name	n	No	Name value is prepended by the Base Name value to form the resource instance name. The resulting name uniquely identifies the Resource Instance. Example: • If Base Name is "/", the Array entry Name of the Resource is {Object}/{Object Instance}/{Resource}/{Resource Instance} • When Base Name is not present, the Array entry Name is the full URI of the requested Resource Instance
	Time	t	No	Representation time relative to the Base Time in seconds for a notification. Required only for historical representations.
	Float Value	v	One value field is mandatory	Value as a JSON float if the Resource data type is Integer, Float, or Time.
	Boolean Value	bv		Value as a JSON Boolean if the Resource data type is Boolean.
	ObjectLink Value	ov		Value as a JSON string if the Resource data type is ObjectLink Format (e.g., "10:03").
	String Value	sv		Value as a JSON string for all other Resource data types. If the Resource data type is opaque the string value holds the Base64 encoded representation of the Resource.

The following is an example of a *bootstrap.ini* file containing security, server, and access control objects, specific to OMA 1.0 servers:

```
{"bn":"/0/0/",
   "e":[
        {"n":"0","sv":"coaps://10.230.20.192:1111"},
        {"n":"1","bv":true},
        {"n":"2","v":0},
        {"n":"10","v":100
        }]}
{"bn":"/0/1/",
   "e":[
        {"n":"1","bv":false},
        {"n":"2","v":3},
        {"n":"10","v":102
        }]}
```

```
{"bn":"/1/1/",
   "e":[
       {"n":"0","v":102},
       {"n":"1","v":50000},
       {"n":"2","v":1},
       {"n":"3","v":60},
       {"n":"5","v":86400},
       {"n":"6","bv":true},
       {"n":"7","sv":"UQS"
       }]}
{"bn":"/2/0",
   "e":[
       {"n":"0","v":1},
       {"n":"1","v":2},
       {"n":"2/102","v":15},
       {"n":"3","v":102
       }]}
```


The following is an example of a *bootstrap.ini* file containing security, server, and access control objects, specific to OMA 1.1 servers:


```
{"bn":"/0/0/",
   "e":[
       {"n":"0","sv":"coap://2002:c023:9c17:c007:5443:ff62:a1f5:2c:5683"},
       {"n":"1","bv":true},
       {"n":"2","v":0},
       {"n":"10","v":100
       }]}
{"bn":"/0/1/",
   "e":[
       {"n":"1","bv":false},
       {"n":"2","v":3},
       {"n":"10","v":102
       }]}
{"bn":"/1/1/",
   "e":[
       {"n":"0","v":102},
       {"n":"1","v":50000},
       {"n":"2","v":1},
       {"n":"3","v":60},
       {"n":"5","v":86400},
       {"n":"6","bv":true},
       {"n":"7","sv":"UN"},
       {"n":"10","ov":"11:65533"},
       {"n":"22","sv":"U"
```

```
        }]}
{"bn":"/2/0",
   "e":[
        {"n":"0","v":1},
        {"n":"1","v":2},
        {"n":"2/102","v":15},
        {"n":"3","v":102
        }]}
```

2.5.1.2. carrier_apn_cfg

carrier_apn_cfg includes details about the carrier-specific APN used for each server.

Table 5: carrier_apn_cfg Items

Parameter	Default Value	Description
APN_NAME	-	APN to be used for the server with SHORT_SERVER_ID1.
APN_CLASS	2	APN class associated with the APN.
SHORT_SERVER_ID1	102	Short server ID associated with the APN.
BS_IF_REG_FAILS	0	If set to 0, LwM2M Client will not perform bootstrapping on registration failure for this SSID. If set to 1, LwM2M Client will perform bootstrapping on registration failure for this SSID.
IP_FAMILY	v4	Supported values: v4, v6 and v4v6. This is the IP family chosen to connect to the server.
BINDINGS	ip_nonip	Supported values: ip, nonip and ip_nonip. This is the binding type to be chosen to specify whether the LwM2M Client can communicate over IP or non-IP.

To successfully register to the network and activate the PDN connection, the corresponding APN should be configured according to the current network. Add a semicolon after the existing APN configuration, and then add the new APN configuration in the next line, as shown below:

```
APN_NAME=carrier_apn1
APN_CLASS=2
BS_IF_REG_FAILS=1
IP_FAMILY=v4
BINDINGS=ip_nonip
SHORT_SERVER_ID1=102
;
```

```
APN_NAME=carrier_apn2
APN_CLASS=3
BS_IF_REG_FAILS=0
IP_FAMILY=v4
BINDINGS=ip_nonip
SHORT_SERVER_ID1=103
;
```

2.5.1.3. lwm2m_cfg

lwm2m_cfg is a LwM2M Client configuration file containing the following configuration information:

Table 6: lwm2m_cfg Items

Parameter	Default Value	Description
APN	-	Default APN to be used if *carrier_apn_cfg* is not available.
IP_FAMILY	v4	Supported values: v4, v6, and v4v6. This is the default IP family chosen to connect to the server if the IP family is not mentioned in *carrier_apn_cfg*.
BINDINGS	ip_nonip	Supported values: ip, nonip and ip_nonip. This is the binding type to be chosen to specify whether the LwM2M Client can communicate over IP or non-IP.
RETRY_TIMEOUT	30	Initial timeout (in seconds) to retry a data call in case of data call failure.
RETRY_EXPONENT_VAL	2	Value by which the data-call retry timeout must be increased exponentially.
MAX_RETRY_TIMEOUT	640	Maximum amount of time (in seconds) for a data call retry.
MAX_NO_RETRIES	5	Maximum number of retry attempts in case of data call failure.
ACK_TIMEOUT	60	Sleep-ACK timeout (in seconds). If there is no activity during the period, the module goes to sleep.
REG_RETRY_TIMEOUT	60	Initial timeout (in seconds) for registration retry in case of registration failure.
REG_RETRY_EXPONENT	2	Value by which the registration retry timeout must be increased exponentially.

REG_RETRY_MAXTIMEOUT	480	Maximum amount of time (in seconds) for a registration retry in case of registration failure.
REG_UPDATE_ON_RECONNECT	0	Determines whether a registration update must be sent on reconnection, such as during reboot with persistence-enabled or when IP address changes: • When set to 1, registration update is sent on reconnection. • When set to 0, registration update is not sent on reconnection.
ROAMING_FEAT	1	Determines LwM2M functionality enablement in roaming: • When set to 1, LwM2M functionality is disabled in roaming. • When set to 0, LwM2M functionality is enabled in roaming.
PER_REGSTATUS_FEAT	0	Determines registration persistence across LwM2M Client reboots: • When set to 1, registration persistence is enabled across LwM2M Client reboots. • When set to 0, registration persistence is disabled across LwM2M Client reboots.
PER_BOOTSTRAP_FEAT	1	Determines bootstrap persistence across LwM2M Client reboots. • When set to 1, bootstrap persistence is enabled across LwM2M Client reboots. • When set to 0, bootstrap persistence is disabled across reboots in the LwM2M Client.
BATTERY_LEVEL_THRESHOLD	20	Battery level less than the threshold mentioned in this configuration value would be considered low battery by the client.
CARRIER_TYPE	0	Enables carrier-specific functionality in the client: 0 Default 1 Verizon
REG_EP_NAME	4	EP name to be used during registration. See OMA specification ***Section 6.3.1 Endpoint Client Name*** for more details. Supported values: 4 IMEI URN 7 IMEI MSISDN URN
BOOTSTRAP_EP_NAME	7	EP name to be used during bootstrap. See OMA specification ***Section 6.3.1 Endpoint Client Name*** for more details. Supported values:

		4 IMEI URN 7 IMEI MSISDN URN
BOOT_UP_SLEEP_TIME	5	Sleep time before an LwM2M Client operation starts.
GPS_MIN_INTERVAL	3000	Interval (in milliseconds) after which GNSS information is fetched periodically.
GPS_MIN_DISTANCE	1	Change in distance (in meters) after which GNSS information is fetched.
MAX_NO_OF_ FACTORY_RESET_ PER_HOUR_BY_SMS	1	Number of factory-reset operations per hour allowed by SMS transport.
MAX_NO_OF_ REG_UPDATE_ PER_HOUR_BY_SMS	1	Number of registration update operations per hour allowed by SMS transport.
WAIT_TIME_TO_ TRIGGER_NEXT_ REG_CYCLE	86400	Hold-off time (in seconds) to trigger the next registration on failure of all registration attempts.
MAX_BOOTSTRAP_WAIT_TIME	0	Hold-off time (in seconds) during which bootstrap procedure is kept on hold, meanwhile extended applications can register objects.
ROOTCA_SUPPORT_MODE	0	Determines whether to use Root CA or domain-issued certificate to validate peer certificate (bootstrap server and standard server) 0 Use domain certificate 3 Use Root CA
SESSION_TIMEOUT	60	Inactivity period (in seconds) after which resumption should be triggered for the uplink operation. OEMs should configure the value to be the same as network address translation (NAT) timeout.
ENABLE_PSM	1	Determines whether LwM2M enables PSM use. 0 Disable 1 Enable
AWAKE_AFTER_PMIN	0	Determines whether the device must go to PSM for every Pmin cycle. 0 By default, the module wakes from Pmin till Pmax. 1 Module goes to PSM for every Pmin cycle until Pmax limit.
STARTUP_MODE	0	Determines whether the LwM2M is initialized with OEM application. 0 By default, LwM2M initialization is preloaded.

		1 LwM2M is initialized with AT commands or DAM application.
VERSION	1.0	Specifies the OMA specification version on which the LwM2M Client is running.

The following example provides an overview of the *lwm2m_cfg* file with the options explained in the table above:

```
{
APN=carrier_apn;
RETRY_TIMEOUT=30;
RETRY_EXPONENT_VAL=2;
MAX_RETRY_TIMEOUT=640;
MAX_NO_RETRIES=5;
ACK_TIMEOUT=30;
REG_RETRY_TIMEOUT=60;
REG_RETRY_EXPONENT=2;
REG_RETRY_MAXTIMEOUT=480;
ROAMING_FEAT=1;
PER_REGSTATUS_FEAT=0;
BATTERY_LEVEL_THRESHOLD=20;
CARRIER_TYPE=0;
REG_EP_NAME=4;
BOOTSTRAP_EP_NAME=7;
BOOT_UP_SLEEP_TIME=10;
STARTUP_MODE=0;
VERSION=1.0;
}
```

2.5.2. Security Mode (DTLS)

Validate the following checkpoints when using Pre-Shared Key security mode:

- PSK files corresponding to bootstrap server or standard server must be present in */datatx/ssl*.
- Files should be in *<ssid>_server.psk* format (e.g., *100_server.psk* for bootstrap server).
- Resource 0/x/2 decides the security mode. See ***Appendix E.1*** of OMA specification for potential values.

A typical PSK file used by the modules contains the following:

- PSK Identity used in transactions with the LwM2M Server.
- PSK Key used in transactions with the LwM2M Server.

An example of credentials required for transactions with the Nokia Motive Bootstrap Server is presented below.

Bootstrap PSK Identity structure is exactly as shown below:

urn:imei-msisdn:<imei_number>-<phone_number>
Example: urn:imei-msisdn:004402460030392-2234235569

The PSK Key used with the bootstrap server is defined as SHA2-256 Digest of the **com.vzwm2m.com** string.

PSK Key = SHA2(com.vzwm2m.com) =
d6160c2e7c90399ee7d207a22611e3d3a87241b0462976b935341d000a91e747

Therefore, the content of *100_server.psk* PSK file for Nokia Motive Bootstrap Server is as follows:

|urn:imei-msisdn:004402460030392-2234235569|:
d6160c2e7c90399ee7d207a22611e3d3a87241b0462976b935341d000a91e747

NOTE

The LwM2M Client accepts any string value as PSK Identity used in a transaction. However, it only accepts hexadecimal value as PSK Key.

3 LwM2M AT Commands

3.1. AT Command Introduction

3.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

3.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 7: Types of AT Commands

Command Type	Syntax	Description
Test Command	**AT+<cmd>=?**	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	**AT+<cmd>?**	Check the current parameter value of the corresponding command.
Write Command	**AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]**	Set user-definable parameter value.
Execution Command	**AT+<cmd>**	Return a specific information parameter or perform a specific action.

3.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

3.3. Description of LwM2M AT Commands

AT commands for configuring and operating an LwM2M Client are introduced in this chapter, whereas a solution for configuring the LwM2M Client with provisioning profiles is provided in ***Chapter 2.5***.

3.3.1. AT+QLWCFG Configure LwM2M Client

The command configures the parameters of LwM2M Client before connecting to the LwM2M Server.

AT+QLWCFG Configure LwM2M Client	
Test Command **AT+QLWCFG=?**	Response **+QLWCFG: "autostart",(**list of supported **<startup_mode>**s**)** **+QLWCFG: "pdpcid",(**range of supported **<client_index>**s**),(**range of supported **<PDP_cid>**s**),(**list of supported **<APN_class>**s**)** **+QLWCFG: "security",(**range of supported **<client_index>**s**),(**range of supported **<shortID>**s**),<server_addr>,(**list of supported **<bootstrap>**s**),(**list of supported **<security_mode>**s**),<PSK_ID>,<PSK_key>** **+QLWCFG: "server",(**range of supported **<client_index>**s**),(**range of supported **<life_time>**s**),(**range of supported **<period_min>**s**),(**range of supported **<period_max>**s**),(**range of supported **<disable_time>**s**),(**list of supported **<storing>**s**),<binding>** **+QLWCFG: "epns",(**list of supported **<type>**s**),<bs_epname>,<reg_epname>** **+QLWCFG: "transcfg",(**range of supported **<retry_timeout>**s**),(**range of supported **<retry_exponent>**s**),(**range of supported **<max_timeout>**s**)** **+QLWCFG: "version",(**list of supported **<client_version>**s**)** **+QLWCFG: "select",(**range of supported

	<server_type>s**),(**list of supported **<is_commercial>**s**)** **+QLWCFG: "host",(**list of supported **<host_index>**s**),(**range of supported **<hostID>**s**),<host_value>** **+QLWCFG: "device",<manufacturer>,<model_no>,<hw_version>,<sw_version>,<fw_version>,<device_type>** **+QLWCFG: "session",(**range of supported **<resume_time>**s**),(**range of supported **<DTLS_timeout>**s**)** **+QLWCFG: "mbsps",<service_code>,<IMEI>,<MSISDN>,<ICCID>,<model>,<MAC>** **OK**
Write Command Query/set the startup mode of LwM2M Client. **AT+QLWCFG="autostart"[,<startup_mode>]**	Response If the optional parameter is omitted, query the current setting. **+QLWCFG: "autostart",<startup_mode>** **OK** If the optional parameter is specified, set the startup mode of LwM2M Client. **OK** Or **ERROR**
Write Command Query/set the APN profile for LwM2M data connection. **AT+QLWCFG="pdpcid"[,<client_index>,<PDP_cid>[,<APN_class>]]**	Response If the optional parameters are omitted, query the current setting. **+QLWCFG: "pdpcid",<client_index>,<PDP_cid>,<APN_class>** **OK** If any of the optional parameters is specified, set the APN profile for LwM2M data connection. **OK** Or **ERROR**
Write Command Query/set the resources of LwM2M security object. **AT+QLWCFG="security"[,<client_index>[,<shortID>,<server_addr>,<bootstrap>,<security_mode>[,<PSK_ID>,<PSK_key>]]]**	Response If the optional parameters are omitted, query the current setting. **+QLWCFG: "security",<client_index>,<shortID>,<server_addr>,<bootstrap>,<security_mode>** **OK** If only **<client_index>** is specified and all other optional parameters are omitted, delete the property data of the specified **<client_index>**.

	OK If **<client_index>** is specified and any of the other optional parameters is specified, set the specified server property. **OK** Or **ERROR**
Write Command Query/set the resources of LwM2M Server object. **AT+QLWCFG="server"[,<client_index>[,<life_time>,<period_min>,<period_max>,<disable_time>,<storing>,<binding>]]**	Response If the optional parameters are omitted, query the current setting. **+QLWCFG: "server",<client_index>,<life_time>,<period_min>,<period_max>,<disable_time>,<storing>,<binding>** **OK** If only **<client_index>** is specified and all other optional parameters are omitted, delete the LwM2M Server attributes of the specified **<client_index>**. **OK** If **<client_index>** is specified and any of the other optional parameters is specified, set the LwM2M Server attributes. **OK** Or **ERROR**
Write Command Query/set the endpoint name that identifies the LwM2M Client on one LwM2M Server. **AT+QLWCFG="epns"[,<type>,<bs_epname>,<reg_epname>]**	Response If the optional parameters are omitted, query the current setting. **+QLWCFG: "epns",<type>,<bs_epname>,<reg_epname>** **OK** If the optional parameters are specified, set the endpoint name. **OK** Or **ERROR**
Write Command Query/set the parameters for bootstrap or registration retry mechanism. **AT+QLWCFG="transcfg"[,<retry_timeout>,<retry_exponent>,<max_timeout>]**	Response If the optional parameters are omitted, query the current setting. **+QLWCFG: "transcfg",<retry_timeout>,<retry_exponent>,<max_timeout>** **OK** If the optional parameters are specified, set the parameters for bootstrap and registration retry mechanism. **OK**

	Or **ERROR**
Write Command Query/set the version of LwM2M Client. **AT+QLWCFG="version"[,<client_version>]**	Response If the optional parameter is omitted, query the current setting. **+QLWCFG: "version",<client_version>** **OK** If the optional parameter is specified, set the version of LwM2M Client. **OK** Or **ERROR**
Write Command Query/set the built-in LwM2M Server information, depending on the specified server type. **AT+QLWCFG="select"[,<server_type>[,<is_commercial>]]**	Response If the optional parameters are omitted, query the current setting. **+QLWCFG: "select",<server_type>,<is_commercial>** **OK** If any of the optional parameters is specified, set the built-in LwM2M Server information. **OK** Or **ERROR**
Write Command Query/set the resource of portfolio object. **AT+QLWCFG="host",<host_index>[,<hostID>[,<host_value>]]**	Response If the optional parameters are omitted, query the current setting of the specified **<host_index>**. If only **<host_value>** is omitted, query the current setting of specified **<host_index>** and **<hostID>**. **[+QLWCFG: "host",<host_index>,<hostID>,<host_value>** **[+QLWCFG: "host",<host_index>,<hostID>,<host_value>** **[...]]]** **OK** If the optional parameters are specified, set the resource of portfolio object. **OK** Or **ERROR**
Write Command Query/set the resource of device object. **AT+QLWCFG="device"[,<manufact**	Response If the optional parameters are omitted, query the current setting. **+QLWCFG: "device",<manufacturer>,<model_no>,<hw_version>,<sw_version>,<fw_version>,<device_type>**

urer>,<model_no>,<hw_version>,<sw_version>,<fw_version>,<device_type>]	**OK** If the optional parameters are specified, set the resource of device object. **OK** Or **ERROR**
Write Command Query/set the session lifetime. **AT+QLWCFG="session"[,<resume_time>,<DTLS_timeout>]**	Response If the optional parameters are omitted, query the current setting. **+QLWCFG: "session",<resume_time>,<DTLS_timeout>** **OK** If the optional parameters are specified, set the session lifetime. **OK** Or **ERROR**
Write Command Query/set the value of parameters required for LG U+ LwM2M Server authentication. **AT+QLWCFG="mbsps"[,<service_code>,<IMEI>,<MSISDN>,<ICCID>,<model>,<MAC>]**	Response If the optional parameters are omitted, query the current setting. **+QLWCFG: "mbsps",<service_code>,<IMEI>,<MSISDN>,<ICCID>,<model>,<MAC>** **OK** If the optional parameters are specified, set the value of parameters required for LG U+ LwM2M Server authentication. **OK** Or **ERROR**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<startup_mode>	Integer type. Enable or disable starting LwM2M automatically when the module powers on. 0 Disable 1 Enable
<client_index>	Integer type. LwM2M Client identifier. Range: 0–5.
<PDP_cid>	Integer type. PDP context ID of LwM2M Client. Range: 1–5. Default value: 1
<APN_class>	Integer type. APN class associated with APN. Supported APN classes: 0, 2, 3, 6,

	7, 10, 11. Default value: 3.
<shortID>	Integer type. Unique identifier of LwM2M Server configured for LwM2M Client. Range: 1–65534.
<server_addr>	String type. LwM2M Server URI. Maximum length: 255 bytes.
<bootstrap>	Integer type. LwM2M Bootstrap Server flag. 0 Standard LwM2M Server 1 LwM2M Bootstrap Server
<security_mode>	Integer type. Encryption method for UDP payload. 0 Pre-Shared Key mode 2 Certificate mode 3 Non-Secure mode
<PSK_ID>	String type. PSK identifier. Only valid when **<security_mode>**=0.
<PSK_key>	String type. PSK secret key. Only valid when **<security_mode>**=0.
<life_time>	Integer type. Registration lifetime. Range: 0–2147483647. Default value: 0. Unit: second.
<period_min>	Integer type. Minimum period of an observation. Range: 0–2147483647. Default value: 0. Unit: second.
<period_max>	Integer type. Maximum period of an observation. Range: 0–2147483647. Default value: 0. Unit: second.
<disable_time>	Integer type. Interval for the next connection after disconnecting from LwM2M Server. Range: 60–2147483647. Default value: 86400. Unit: second.
<storing>	Integer type. LwM2M Client stores “Notify” operations to LwM2M Server or discards them when LwM2M Server account is disabled or LwM2M Client is offline. 0 LwM2M Client stores “Notify” operations 1 LwM2M Client discards “Notify” operations
<binding>	String type. Transport binding mode configured for LwM2M Client. See ***Table 8***.
<type>	Integer type. Format of endpoint client name. 0 String type - Specify endpoint client name with string format. 1 Integer type - Value of **<bs_epname>** and **<reg_epname>** should be as in ***Table 9***.
<bs_epname>	Endpoint client name for identifying bootstrap server. Format: determined by **<type>**. See ***Table 9***.
<reg_epname>	Endpoint client name for identifying standard server. Format: determined by **<type>**. See ***Table 9***.
<retry_timeout>	Integer type. Initial timeout value for bootstrap or registration retry upon failure. Range: 0–2147483647. Default value: 60. Unit: second.
<retry_exponent>	Integer type. Value by which the timeout for bootstrap or registration retries must be increased exponentially. Range: 0–2147483647. Default value: 2.
<max_timeout>	Integer type. Maximum timeout for all bootstrap or registration retries upon failure. Range: 0–2147483647. Default value: 480. Unit: second.
<client_version>	Integer type. LwM2M protocol version. 0 LwM2M protocol version 1.0 1 LwM2M protocol version 1.1

<server_type>	Integer type. Built-in LwM2M Server details for network carrier. 0 Factory mode, used to reset all settings of current LwM2M Client 1 Verizon built-in LwM2M Server information 2 AT&T built-in LwM2M Server information 3 T-Mobile built-in LwM2M Server information 4 Telstra built-in LwM2M Server information 5 LG U+ built-in LwM2M Server information 6 NTT DOCOMO built-in LwM2M Server information
<is_commercial >	Integer type. Whether preloaded LwM2M Server details are for commercial use. 0 Staging Server 1 Production Server
<host_index>	Integer type. Instance ID of host device. Supported values: 0 and 1.
<hostID>	Integer type. Resource ID associated with host device instance. 0 Host device ID 1 Host device manufacturer 2 Host device model number 3 Host device software version
<host_value>	String type. Host device information.
<manufacturer>	String type. Device manufacturer.
<model_no>	String type. Device model number.
<hw_version>	String type. Device hardware version.
<sw_version>	String type. Device software version.
<fw_version>	String type. Device firmware version.
<device_type>	String type. Device type name.
<DTLS_timeout>	Integer type. DTLS session timeout. Range: 0–2147483647. Default value: 0. Unit: second.
<resume_time>	Integer type. DTLS session resumption timeout. Range: 0–86400. Default value: 86400. Unit: second.
<service_code>	String type. Service code.
<IMEI>	String type. Device IMEI number.
<MSISDN>	String type. MSISDN number.
<ICCID>	String type. ICCID number.
<model>	String type. Device model name.
<MAC>	String type. Device MAC address.

Table 8: <binding> Arguments

<binding>	Description
"U"	UDP. LwM2M Server expects LwM2M Client to be accessible via UDP binding at any time.
"UQ"	UDP with Queue Mode. LwM2M Server MUST queue all requests to LwM2M Client, and send requests via UDP when the LwM2M Client is on-line.
"S"	SMS. LwM2M Server expects LwM2M Client to be accessible via SMS binding at any time.
"SQ"	SMS with Queue Mode. LwM2M Server MUST queue all requests to LwM2M Client, and send requests via SMS when the LwM2M Client is on-line.
"US"	UDP and SMS. LwM2M Server expects LwM2M Client to be accessible via UDP binding and SMS binding at any time.
"UQS"	UDP with Queue Mode and SMS. LwM2M Server MUST queue all requests to LwM2M Client, and send requests via UDP when the LwM2M Client is on-line. The LwM2M Server expects the LwM2M Client to be accessible via SMS binding at any time.
"N"	Non-IP. LwM2M Server MUST send requests to LwM2M Client using Non-IP binding. The LwM2M Client MUST send the response to such requests over the Non-IP binding.

If **<type>**=1, the valid values of **<bs_epname>** and **<reg_epname>** are listed in the table below.

Table 9: <bs_epname> and <reg_epname> Arguments

<bs_epname>/ <reg_epname>	Description
4	Identify a module using an International Mobile Equipment Identity (see *3GPP TS 23.003*). The IMEI URN specifies a valid, 15-digit IMEI. URN format: "urn:imei:<IMEI>".
5	Identify a module using an Electronic Serial Number. ESN URN specifies a valid, 8-digit ESN. URN format: "urn:esn:<ESN>".
6	Identify a module using a Mobile Equipment Identifier. MEID URN specifies a valid, 14-digit MEID. URN format: "urn:meid:<MEID>".
7	Identify a module using a combination of International Mobile Equipment Identity (see *3GPP TS 23.003*), and MSISDN. Both IMEI and MSISDN are 15 digit numbers. URN format: "urn:imei-msisdn:<IMEI>-<MSISDN>".
8	Identify a module using a combination of International Mobile Equipment Identity (see *3GPP TS 23.003*), and IMSI. Both IMEI and IMSI are 15 digit numbers. URN format: "urn:imei-imsi:<IMEI>-<IMSI>".
9	Identify a module using a combination of International Mobile Equipment Identity (see *3GPP TS 23.003*), and IMSI. Both IMEI and IMSI are 15 digit numbers. URN

	format: "imei-imsi:<IMEI>-<IMSI>".

NOTE

1. LwM2M security and server objects must be initialized before starting up LwM2M Client. **AT+QLWCFG="security"** must be executed before you configure the resource values of LwM2M Server object and parameters of LwM2M data session.
2. Transport binding mode "N" is only supported under LwM2M protocol version 1.1.

3.3.2. AT+QLWSVC Manage LwM2M Session

The command operates a LwM2M Client and manages the flow of LwM2M session.

AT+QLWSVC Manage LwM2M Session	
Test Command **AT+QLWSVC=?**	Response **+QLWSVC: "reg",(**list of supported **<mode>**s**)** **+QLWSVC: "dereg",(**list of supported **<method>**s**)** **+QLWSVC: "lifetime",(**range of supported **<shortID>**s**),(**range of supported **<life_time>**s**)** **+QLWSVC: "update",(**range of supported **<shortID>**s**),(**list of supported **<with_object>**s**)** **+QLWSVC: "state",(**range of supported **<client_index>**s**)** **+QLWSVC: "uldata",(**range of supported **<shortID>**s**),<URI>,(**range of supported **<data_len>**s**),<data>,(**list of supported **<data_type>**s**),(**list of supported **<message_type>**s**)** **OK**
Write Command Initiate registration with LwM2M Server. **AT+QLWSVC="reg"[,<mode>]**	Response **OK** **+QLWREG: <result>** If there is an error: **+CME ERROR: <cme_err>**
Write Command Deregister a LwM2M Client from the LwM2M Server. **AT+QLWSVC="dereg"[,<method>]**	Response **OK** **+QLWDEREG: <result>** If there is an error: **ERROR**

Write Command Query/update the lifetime of a specific LwM2M Server. **AT+QLWSVC="lifetime",<shortID>[,<life_time>]**	Response If the optional parameter is omitted, query the lifetime value of a specific LwM2M Server. **+QLWSVC: "lifetime",<shortID>,<life_time>** **OK** If the optional parameter is specified, update the lifetime of a specific LwM2M Server. **OK** Or **ERROR**
Write Command Send a registration update to a specific LwM2M Server. **AT+QLWSVC="update",<shortID>[,<with_object>]**	Response If the registration update message is sent successfully: **OK** If there is any error: **ERROR**
Write Command Query LwM2M Client registration state. **AT+QLWSVC="state"[,<client_index>]**	Response **+QLWSVC: "state",<client_index>,<shortID>,<server_addr>,<bootstrap>,<state>** **[…]** **OK**
Write Command Send application data over LwM2M session. **AT+QLWSVC="uldata",<shortID>,<URI>,<data_len>,<data>[,<data_type>[,<message_type>]]**	Response **OK** If there is any error: **ERROR**
Maximum Response Time	300 ms
Characteristics	/

Parameter

<mode>	Integer type. Registration mode. 0 LwM2M Client initiates “registration” after the module reboots. 1 LwM2M Client initiates “registration update” after the module reboots.
<method>	Integer type. Deregistration method. 0 Deregistration 1 Factory reset
<shortID>	Integer type. Unique identifier of LwM2M Server configured for LwM2M Client.

	Range: 1–65534.
<life_time>	Integer type. Registration lifetime. Range: 0–2147483647. Unit: second.
<with_object>	Integer type. Whether the current registration update message contains an object list. 0 Send the registration update message with NULL payload. 1 Send the registration update message with object list.
<client_index>	Integer type. LwM2M Client identifier. Range: 0–5.
<server_addr>	String type. LwM2M Server URI.
<bootstrap>	Integer type. LwM2M Bootstrap Server flag. 0 Standard LwM2M Server 1 LwM2M Bootstrap Server
<state>	Integer type. Registration state code of LwM2M Client. See **<state_code>** in ***Chapter 3.4.1***.
<URI>	String type. URI format: /<Object ID>/<Instance ID>/<Resource ID>.
<data_len>	Integer type. Application data length. Range: 1–1460. Unit: byte.
<data>	String type or hexadecimal type. Application data to be sent.
<data_type>	Integer type. Application data type. 0 String type. Send application data in string format. 1 Hexadecimal type. Send application data in hexadecimal format.
<message_type>	Integer type. Message type of application data to be sent. 0 Confirmable 1 Non-confirmable
<result>	Integer type. 0 indicates a successful operation. Other values indicate failure. 0 Operation success 1 Operation failure 2 Invalid parameter 3 No memory 4 Failure as another parallel operation started 5 Operation is not allowed
<cme_err>	Integer type. Error code of AT command operation. 701 Unknown error 702 Command has been executed 703 Invalid parameter 704 LwM2M task is not initiated yet 705 Operation is not allowed 706 No memory

NOTE

If **AT+QLWSVC="lifetime"** is executed, a "registration update" CoAP message containing lifetime value is sent to LwM2M Server. If the module fails to send this CoAP message or receives any response from LwM2M Server, the corresponding event notification in the below format is reported:
+QLWURC: "event",<shortID>,<state_code>,<state_string>

See ***Chapter 3.4.1*** for URC details.

3.4. Description of LwM2M URCs

URCs of LwM2M AT commands are reported to the host when any internal event happens on LwM2M Client, including bootstrapping, registration, observation, and incoming application data.

3.4.1. +QLWURC: "event" LwM2M Client State Change Notification

This URC is reported when the LwM2M Client state changes during bootstrapping or registration; for instance, when the lifetime of a specific LwM2M Server is updated, or when you send a registration update to a specific LwM2M Server.

+QLWURC: "event" LwM2M Client State Change Notification	
+QLWURC: "event",<shortID>,<state_code>,<state_string>	Notify the current state of LwM2M Client.

Parameter

<shortID>	Integer type. Unique identifier of the LwM2M Server configured for the LwM2M Client. Range: 0–65534. The value 0 is only used for reporting "GENERIC_FAILED" and "FACTORY_RESET" related notifications.
<state_code>	Integer type. Current state code of LwM2M Client. See ***Table 10***.
<state_string>	String type. Readable string associated with LwM2M Client current state code. See ***Table 10***.

Table 10: <state_code> and <state_string> Arguments

<state_code>	<state_string>	Description
0	INITIAL	LwM2M Client not registered or bootstrap not started.
1	REG_PENDING	Registration pending.
2	REG_READY	Registered on LwM2M Server successfully.
3	REG_FAILED	Last registration failed.
4	UPDATE_PENDING	Registration update pending due to LwM2M Client in sleep state.

5	FACTORY_RESET	LwM2M Client in bootstrap mode due to running hold-off timer.
6	BS_START	LwM2M Client initiates a bootstrap request.
7	BS_PENDING	Bootstrap ongoing.
8	BS_HOLDOFF	LwM2M Client is waiting for the pass-through object initialization to complete and the "bootstrap request" message has not yet been sent.
9	BS_FINISH	Bootstrap finished.
10	BS_FAILED	Bootstrap failed.
11	REG_CONN_RETRY	Data call connection retry on failure during registration phase.
12	REG_RETRY	Registration retry on failure.
13	BS_CONN_RETRY	Data call connection retry on failure during bootstrapping phase.
14	BS_RETRY	Bootstrap retry on failure.
15	GENERIC_FAILED	Generic error usually caused by invalid parameters, populated endpoint name for bootstrap, or failed registration and failed data call start.
16	REG_UPDATE	Registration update sent successfully.
17	UPDATE_FAILED	Registration update sending failed.
18	DEREG_START	Trigger deregistration from LwM2M Server.
19	DEREG_FINISH	Deregistered from LwM2M Server successfully.

3.4.2. +QLWURC: "observe" Observe Request Indication

When specific resources are observed by LwM2M Server, this URC is reported.

+QLWURC: "observe" Observe Request Indication	
+QLWURC: "observe",<shortID><URI>,<messageID>,<token>,<flag>	Notify that there is an Observe Request from server.

Parameter

<shortID>	Integer type. Unique identifier of the LwM2M Server configured for the LwM2M Client. Range: 1–65534.
<URI>	String type. URI path of Observe operation. URI format: /<Object ID>/<Instance ID>/<Resource ID>.
<messageID>	Integer type. Message ID of Observer CoAP message.
<token>	Integer type. Token ID of Observe CoAP message.
<flag>	Integer type. Observe flag. 0 Observe 1 Cancel Observe

3.4.3. +QLWURC: "uldata" Application Data Delivery Result Notification

When the application data are sent in confirmable message type, this URC is reported after receiving any response from the LwM2M Server.

+QLWURC: "uldata" Application Data Delivery Result Notification	
+QLWURC: "uldata",<shortID>,<URI>,<messageID>,<token>,<response_code>	Notify the application data sending status returned by the server.

Parameter

<shortID>	Integer type. Unique identifier of the LwM2M Server configured for the LwM2M Client. Range: 1–65534.
<URI>	String type. URI format: /<Object ID>/<Instance ID>/<Resource ID>.
<messageID>	Integer type. Message ID of CoAP message.
<token>	Integer type. Token ID of CoAP message.
<response_code>	Integer type. Response code used to indicate the result of application data delivery. 0 Application data with confirmable message type sent successfully -1 Received error response code from server -2 Received NULL response from server -3 Received unsupported or invalid response message from server

3.4.4. +QLWURC: "recv" Incoming Data Indication

When the LwM2M Client has received downlink data from the server, this URC is reported.

+QLWURC: "recv" Incoming Data Indication	
+QLWURC: "recv",<URI>,<data_type>,<data_len>,<CR><LF><data>	Notify incoming data in direct push mode.

Parameter

<URI>	String type. URI format: /<Object ID>/<Instance ID>/<Resource ID>.
<data_type>	Integer type. Content type of incoming downlink data. 4 Resource value type is string. 5 Resource value type is opaque.
<data_len>	Integer type. Length of incoming downlink data. Unit: byte. Maximum number of bytes that can be received at a time: 1024. Unit: byte
<data>	Incoming downlink data.

4 Examples

4.1. Configure LwM2M Client with Provisioning Profiles*

1) Use *bootstrap.ini* file to configure LwM2M security object and server object information for Leshan Server.

- Factory Bootstrap (in Non-Secure mode)

```
{"bn":"/0/1/",
   "e":[
        {"n":"0","sv":"coap://leshan.eclipseprojects.io:5683"},
        {"n":"1","bv":false},
        {"n":"2","v":3},
        {"n":"10","v":102
        }]}
{"bn":"/1/1/",
   "e":[
        {"n":"0","v":102},
        {"n":"1","v":50000},
        {"n":"2","v":1},
        {"n":"3","v":60},
        {"n":"5","v":86400},
        {"n":"6","bv":true},
        {"n":"7","sv":"UQS"
        }]}
```

- Factory Bootstrap (in Pre-Shared Key mode)

```
{"bn":"/0/1/",
   "e":[
        {"n":"0","sv":"coaps://leshan.eclipseprojects.io:5684"},
        {"n":"1","bv":false},
        {"n":"2","v":0},
        {"n":"10","v":102
        }]}
{"bn":"/1/1/",
   "e":[
```

```
{"n":"0","v":102},
{"n":"1","v":50000},
{"n":"2","v":1},
{"n":"3","v":60},
{"n":"5","v":86400},
{"n":"6","bv":true},
{"n":"7","sv":"UQS"
}]}
```


An additional PSK key file must be loaded into */datatx/ssl* folder in case of Pre-Shared Key mode.

2) Confirm with network carrier and configure proper APN and APN class in *carrier_apn_cfg* and *lwm2m_cfg* files. For details, see ***Chapter 0*** and ***Chapter 0***.

3) Put *lwm2m_app_autostart* auto startup script into */datatx* and reboot the module. When the module registers on Leshan Server successfully, the URCs listed below will be reported.

```
+QLWURC: "event",102,0,"INITIAL"

+QLWURC: "event",102,2,"REG_READY"
```

4.2. Configure LwM2M Client with AT Command

4.2.1. Configure in Client-Initiated Bootstrap Mode

```
//Configure LwM2M security object in Pre-Shared Key mode.
AT+QLWCFG="security",0,111,"coaps://leshan.eclipseprojects.io:5784",1,0,"urn:imei:864475040
090705","313233343536373839"
OK

//Query the current resource value of LwM2M security object.
AT+QLWCFG="security"
+QLWCFG: "security",0,111,"coaps://leshan.eclipseprojects.io:5784",1,0

OK
AT+QLWCFG="server",0,60,0,50,60,1,"UQS"    //Configure the LwM2M Server object.
OK
AT+QLWCFG="server"
+QLWCFG: "server",0,60,0,50,60,1,"UQS"

OK
AT+QLWSVC="reg"                            //Initiate a bootstrap request to the server.
```

```
OK

+QLWREG: 0

+QLWURC: "event",111,6,"BS_START"

+QLWURC: "event",111,9,"BS_FINISH"

+QLWURC: "event",123,0,"INITIAL"

+QLWURC: "event",123,2,"REG_READY"

//The LwM2M Client can execute the following commands only after registering on the server.
AT+QLWSVC="lifetime",123,60                   //Update registration lifetime to 60 s.
OK

+QLWURC: "event",123,16,"REG_UPDATE"
AT+QLWSVC="update",123,0                      //Send "registration update" message to LwM2M
                                              Server. If the module is sleeping, execute this
                                              command to wake up the module.
OK

+QLWURC: "event",123,16,"REG_UPDATE"

//Modify "Device Object" information for the host.
AT+QLWCFG: "device","Quectel","BG770A","R1.1","33","BG770AGLAAR01A03","MODEM"
OK
AT+QLWSVC="dereg"                             //LwM2M Client is deregistering from LwM2M Server.
OK

+QLWURC: "event",123,19,"DEREG_FINISH" //LwM2M Client deregistered from server successfully.

+QLWDEREG: 0
```

4.2.2. Configure in Factory Bootstrap Mode

```
//Configure LwM2M security object in Non-Secure mode.
AT+QLWCFG="security",0,100,"coap://leshan.eclipseprojects.io:5683",0,3
OK
AT+QLWCFG="server",0,60,0,50,60,1,"UQS"        //Configure LwM2M Server object.
OK
AT+QLWSVC="reg"                                //Initiate a registration request to the server.
OK
```

```
+QLWREG: 0

+QLWURC: "event",100,0,"INITIAL"

+QLWURC: "event",100,2,"REG_READY"

//The LwM2M Client can execute the following commands only after registering on the server.
AT+QLWSVC="lifetime",100,60          //Update registration lifetime to 60 s.
OK

+QLWURC: "event",100,16,"REG_UPDATE"
AT+QLWSVC="update",100,0             //Send "registration update" message to LwM2M
                                     Server, If the module is sleeping, execute this
                                     command to wake up the module.
OK

+QLWURC: "event",100,16,"REG_UPDATE"
AT+QLWSVC="dereg",1                  //Factory reset.
OK

+QLWURC: "event",0,5,"FACTORY_RESET"

+QLWURC: "event",100,18,"DEREG_START"

+QLWURC: "event",100,19,"DEREG_FINISH"

+QLWDEREG: 0
```

5 Appendix References

Table 11: Terms and Abbreviations

Abbreviation	Description
ACK	Acknowledgement
ACL	Access Control List
APN	Access Point Name
APPS	Application Subsystem
CoAP	Constrained Application Protocol
DAM	Downloadable Application Module
DTLS	Datagram Transport Layer Security
EP	Endpoint
FOTA	Firmware Over-The-Air
GNSS	Global Navigation Satellite System
ICCID	Integrated Circuit Card Identifier
IMEI	International Mobile Equipment Identity
IoT	Internet of Things
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
JSON	JavaScript Object Notation
LwM2M	Lightweight Machine to Machine
MAC	Medium Access Control

MSISDN	Mobile Subscriber Integrated Services Digital Network
NAT	Network Address Translation
OEM	Original Equipment Manufacturer
OMA	Open Mobile Alliance
PDN	Public Data Network
PDP	Packet Data Protocol
Pmax	Maximum Period
Pmin	Minimum Period
PSK	Pre-Shared Key
PSM	Power Saving Mode
SenML	Sensor Measurement Lists
SMS	Short Message Service
SSID	Service Set Identifier
SSL	Secure Sockets Layer
UDP	User Datagram Protocol
UI	User Interface
URC	Unsolicited Result Code
URI	Uniform Resource Identifier
URN	Uniform Resource Name